Environmental Reviews

Review

Cigarette butts in the environment: a growing global threat?

Nirhvana Felipe da Silva[a], **Maria Christina Barbosa de Araújo**[b], **and Jacqueline Santos Silva-Cavalcanti**[a,c]

[a]Graduate Program in Biodiversity, Federal Rural University of Pernambuco, Manuel de Medeiros, 97 - Dois Irmãos, Recife, Pernambuco, Brazil; [b]Department of Oceanography and Limnology, Federal University of Rio Grande do Norte, Governador Silvio Pedroza, 1 - Areia Preta, Natal, Rio Grande do Norte, Brazil; [c]Department of Biology, Federal Rural University of Pernambuco, Manuel de Medeiros, 97 - Dois Irmãos, Recife, Pernambuco, Brazil

Corresponding author: **Jacqueline Santos Silva-Cavalcanti** (email: jacque_ss@hotmail.com)

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Abstract

Cigarette butts (CBs) are composed of cellulose acetate and are a significant source of anthropogenic waste. More than 4 trillion CBs are improperly discarded in natural and urban environments, resulting in the contamination of a variety of ecosystems. The goal of the present study was to obtain information regarding environmental contamination of CBs through a comprehensive systematic review. A literature review was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses method. "Cigarette butt" and "cigarette filter" were used as keywords in searches in Google Scholar, Scopus, PubMed, ScienceDirect, and SpringerLink databases, where the abstracts were separated, organized, and analysed using IRaMuTeQ software. The review identified 116 articles published in 23 countries, with publication growth observed over the years. Through descending hierarchical classification, two groups and four classes were recognized, whereby different terminologies were specified by factorial correspondence and similarity analyses. The four classes were categorized as follows: (1) ecotoxicological studies, with information about the lethal and sublethal effects of CBs on different organisms; (2) public policies, with discussion pertaining to the problem and possible measures and actions aimed at reducing CB contamination; (3) contamination of public areas, with studies addressing the potential dispersion of this material in the environment; and (4) physicochemical aspects, with evidence of the potential for contamination caused by the components contained in the cigarette filters. However, despite an increasing number of publications over the years and a variety of studies regarding the environmental effects of CBs, there is still an absence of information within each class, requiring further research.

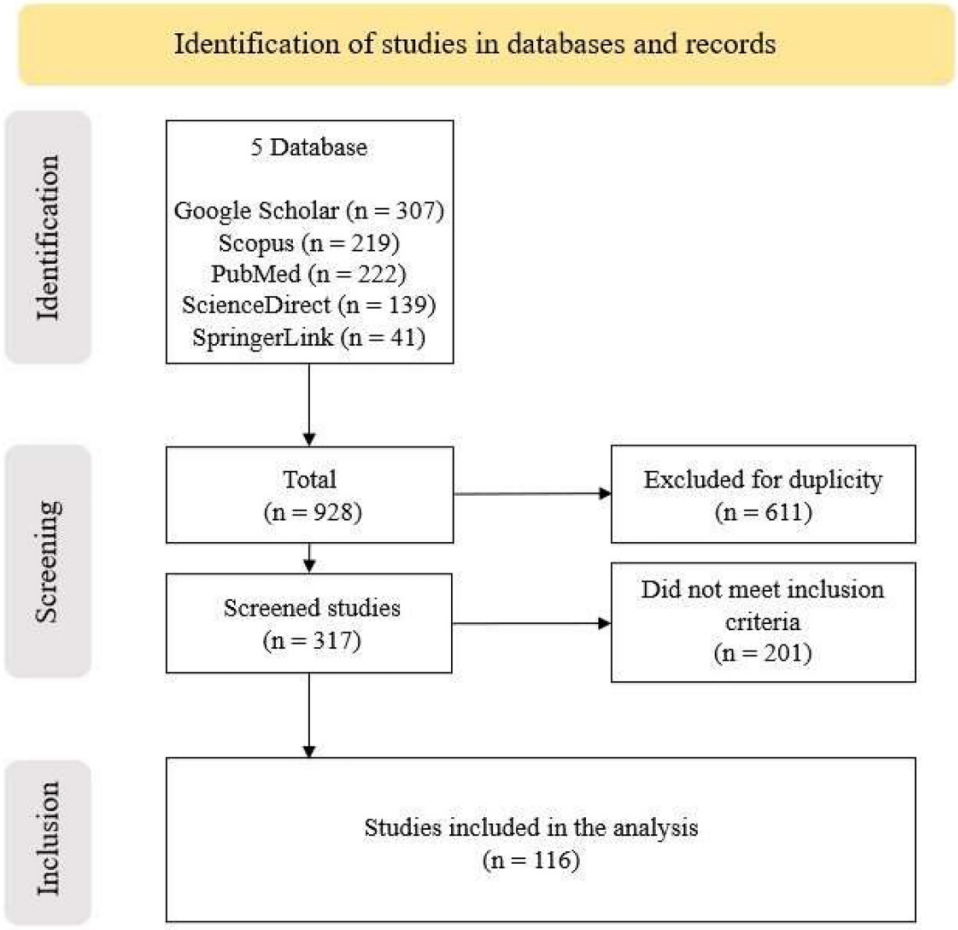
Key words: scientometrics, solid waste, cellulose acetate, nanofibers

1. Environmental contamination from cigarette butts: scientometric analysis

Natural environments are littered with garbage worldwide. This problem affects both inhabited places (populous or not) and remote, isolated regions (Asensio-Montesinos et al. 2020; Lozano-Rivas et al. 2020; Grillo and Mello 2021; Lima et al. 2021; Ramos et al. 2021). In marine environments, this problem has been identified as one of the main factors responsible for biodiversity degradation (Fauziah et al. 2021). Although garbage is composed of a wide variety of waste types, and there is a demonstrated prevalence of plastic items (Dobaradaran et al. 2017, 2018*a*; Akhbarizadeh et al. 2021*a*; De-la-Torre et al. 2022*a*, *b*), an item of synthetic composition, i.e., cigarette butts (CBs), has stood out for its global occurrence and quantitative magnitude. CBs are considered the most common form of personal waste worldwide (Novotny and Slaughter 2014; Dobaradaran et al. 2018*b*). For example, surveys performed between 2012 and 2019 by Ocean Conservancy indicated that this type of compound is the most abundant type of marine litter (Ocean Conservancy 2022).

It has been estimated that 5.5 trillion cigarettes are consumed each year worldwide, and 4.95 trillion CBs are improperly discarded into the environment (Novotny and Slaughter 2014). Places with greater human activity have a higher number of items. Tourist activity and proximity to urban centers are responsible for the majority of CBs in marine environments (Araújo and Costa 2007; Ribeiro et al. 2021). Approximately two-thirds of all CBs are discarded by smokers in public places (Novotny and Slaughter 2014). High concentrations of CBs can be observed around beaches (Araújo and Costa 2021) or in regions where cigarettes are sold (Marah and Novotny 2011). In addition, CBs remain for long periods of time in these places, as they are not efficiently removed by cleaning services, causing greater damage to public health and the environment (Silva et al. 2008).

Fig. 1. Flowchart of steps for selecting articles using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses method.



Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Cigarette filters are composed of cellulose acetate, a synthetic polymer derived from cellulose (Araújo and Costa 2019), characterized by high durability and slow decomposition rates. Polymer degradation can lead to the accumulation of nanoparticles in soils and the ocean (El Hadri et al. 2021; Oliva et al. 2021). This material contains compacted fibers and can absorb compounds such as nicotine, lead, copper, chromium, cadmium, additives, and pesticide residue that are present in tobacco. Consequently, soils and aquifers can become contaminated through leaching (Lee and Lee 2015).

Cigarettes and their waste generally have a physical structure composed of filters, tobacco, and paper. Studies on environmental contamination have included a broad classification of cigarette waste that varies from paper, plastic, mixed composition, or "others", making it difficult to standardize the handling of cigarette residues and formulate management plans. Currently, the literature on the environmental damage caused by CBs disposal is still scarce and requires special attention. The goal of the present study was to better understand the environmental impacts of CBs by performing lexicometric analyses of contamination from CBs via comprehensive scientometric analysis.

2. Approach

2.1. Bibliographic search

To assess the environmental impacts associated with CBs, a literature search was performed in June 2022 using the following platforms: Google Scholar, Scopus, PubMed, ScienceDirect, and SpringerLink. The keywords "cigarette butt" and "cigarette filter" were used, without any temporary intermittence. The review assessed publications in English and their respective references for possible additional citations. Subsequently, publications were sorted based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses pattern, outlining the stages using a flowchart as shown in Fig. 1.

2.2. Exclusion and inclusion criteria

Three procedures were used for exclusion and inclusion criteria: (1) removal of duplicate articles, (2) selection of publications that meet the objectives and purpose of the present study (CBs and their impacts on the environment), and (3) inclusion limited to articles, with the removal

Table 1. Expressions used as variations and synonyms for the term "cigarette butts" found in the screened abstracts, along with their values in percentage (%) and number (*n*) observed in abstracts of other articles in the databases assessed.

Words/expressions	Concept	Screened studies (%)	Studies in the databases (*n*)
Material			
CB/CBs	Used as diminutive for "cigarette butt"	39.08%	134
Butt waste	Cigarette-derived waste, used to describe butt residues or butt fibers	13.79%	4
Cigarette waste	Cigarette-derived waste, used to describe tobacco, paper, or butt residues	3.44%	2
Filtered cigarette	"Filtered cigarettes" or "cigarettes with filters"	2.29%	10
Chemical composition			
Cellulose filter	Material from cigarette filters	2.29%	50
Cellulose acetate filter	Fibers from cigarette butts composed of cellulose acetate	2.29%	3
Cellulose fibers	Cellulose fibers resulting from cigarette butts	1.14%	832
Cellulose microfibers/nanofibers	Cellulose microfibers or nanofibers from cigarette butts	1.14%	629

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

of proceedings, conference abstracts, editorials, and book chapters.

Initially, 928 articles were found in the databases, of which 611 were duplicates and 201 were deemed not to meet the criteria and objectives of the present study (e.g., did not provide information on environmental perspectives). Ultimately, 116 articles that met all the inclusion criteria were selected for the study (Fig. 1).

After screening and selecting the studies, content analyses and a lexicometric study of the abstracts were performed, grouping them based on the years and countries of publication. Descending hierarchical classification (DHC), correspondence factor analysis (CFA), and similarity analysis were performed using IRaMuTeQ (version 0.7, alpha 2) software for multivariate textual ordering.

2.3. Analysis of the studies

In DHC (Reinert 1987), the formation of classes is performed from a dendrogram according to the arrangement of the most expressive (representative) words of each class. This procedure allows the description of each word, aiming at greater understanding and organisation of the information provided in the studies with respect to the environmental implications of CBs. CFA (Benzécri 1973) provides the disposition of classes in a factorial plan, where it is possible to determine the proximity and distance of the classes, revealing the connectivity of information and discussion regarding CBs and environmental damage. Meanwhile, similarity analysis (based on graph theory) establishes the relationship between words from a branching pattern. Furthermore, it assesses the existence of links and coexistence between words, allowing for the observation of the relationships between the different terms used in publications about CBs and the environmental damage they cause.

3. Discussion

Despite the variations and synonyms for the word "butt" found during the present review, it was possible to observe the predominance of the term "cigarette butt" in the 116 articles assessed (Table 1).

In addition to the different terms and words used to describe CBs, the connections between them could also be observed. The words used as keywords in the literature search appeared interconnected in the similarity analysis (Fig. 2), making it possible to assess the emergence of 73 branches, with 45 (61.64%) for "cigarette" and 28 (38.36%) for "butt". These values can be associated with the expressive number of synonyms and variations that were used in the texts, with the predominance of the word "cigarette". According to the similarity analysis, it was also possible to determine the contribution of words in the formation of the tree (in which larger words had greater frequency and contribution), with uniformity between the terms. However, the ramifications were limited, reiterating the need for more studies on solving CB issues. Nevertheless, it was possible to observe that based on the vocabulary of the articles, there was concern regarding CBs composition. The emphasis was on its elements, categorisation as a toxic component, effects on the environment (with emphasis on marine environments), and human action related to the disposal of CBs.

The similarities and links of the content occurred over time, with a correlation between the studies that addressed CBs from an environmental perspective and the discussion on its impacts on biodiversity. The relationship between one of the keywords and environmental damage was first mentioned in the title of a 2006 publication (Fig. 3) that evaluated the toxicity of 19 cigarette filters in different aquatic organisms (Micevska et al. 2006) and highlighted the severe consequences of the presence of this material in the environment.

Another review addressing environmental pollution from CBs indicated the low biodegradability of CBs, as well as its

Fig. 2. Similarity analysis of the assessed abstracts.

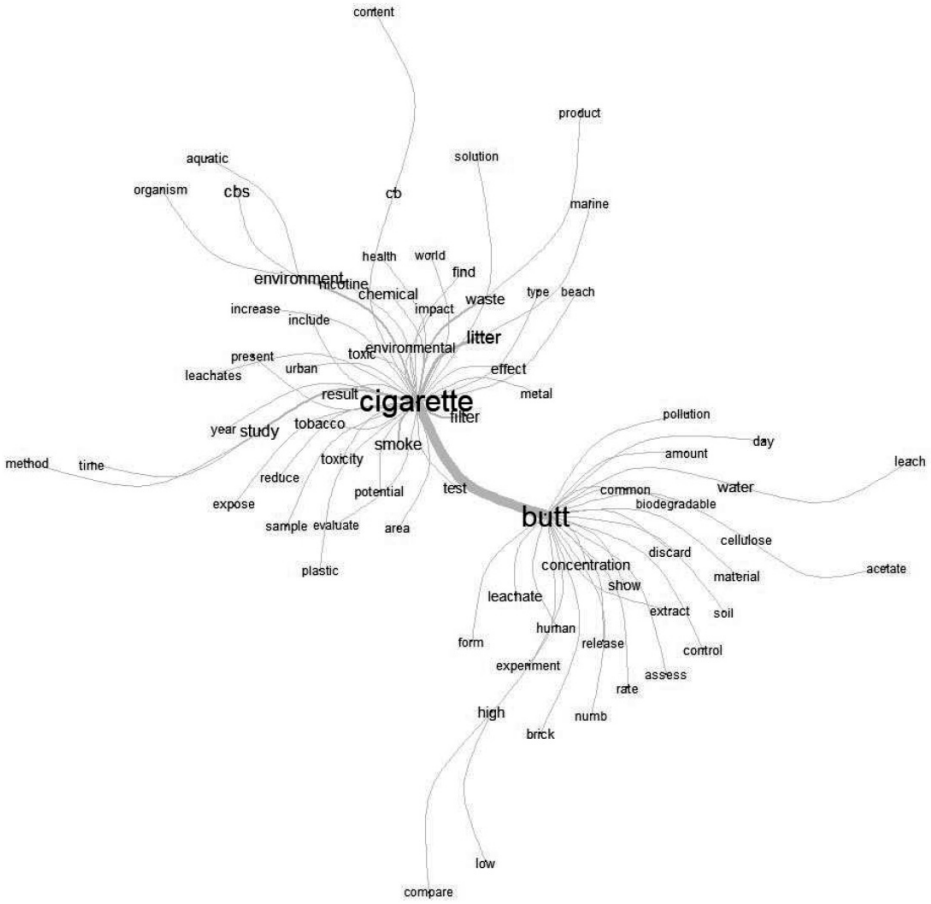
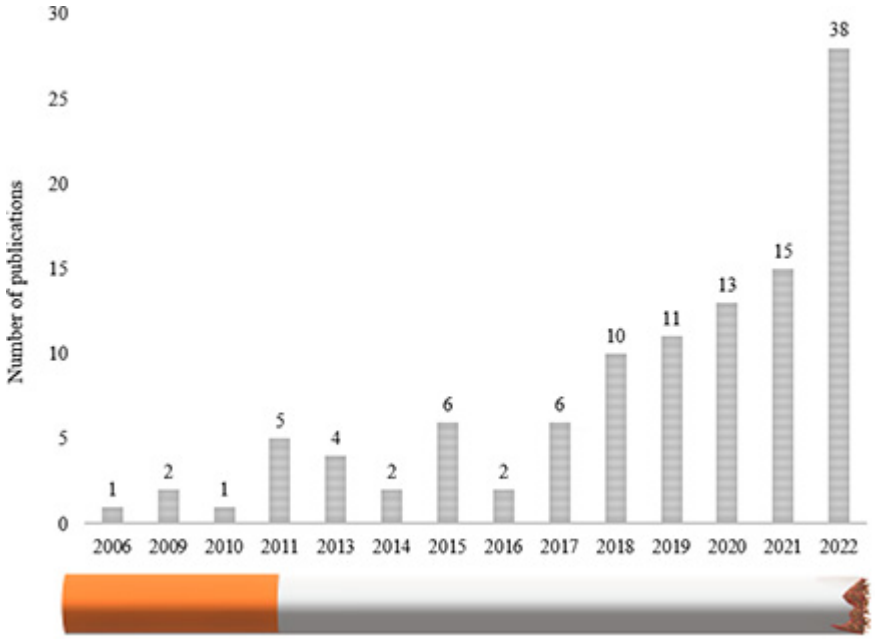


Fig. 3. Number of articles published addressing environmental impacts caused by cigarette butts between 2006 and 2022.



Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

high dispersion and toxicity (Kadir and Sarani 2015). From 2015 to 2021, 63 articles were compiled, or 72.41% of all content evaluated in this study, providing evidence of the timeliness of the subject. This increase may be associated with the high worldwide recognition of similar topics such as microplastics, mesoplastics, and marine debris. These subjects include CBs, as studies assessing waste disposal have often highlighted the presence of the contaminant in alarming numbers (Becherucci et al. 2017; Marin et al. 2019; Thiel et al. 2021). Furthermore, according to Ocean Conservancy (2022), the global quantity of registered CBs increased from 1 910 777 items in 2012 to 5 716 331 in 2019, which may influence the expansion of knowledge by explaining the high dispersion capacity of CBs registered over the years.

Between 2006 and 2022, publications reporting environmental impacts associated with the inappropriate disposal of CBs were recorded on five continents (Fig. 4). The country with the highest number of publications on the subject was the United States ($n = 21$), followed by Iran ($n = 18$), Brazil ($n = 16$), Italy ($n = 15$), France ($n = 10$), Spain and Malaysia ($n = 5$), Australia and the United Kingdom ($n = 3$), Greece, India, Germany, Nigeria, and Turkey ($n = 2$). Mexico, Taiwan, Cyprus, South Africa, Finland, Lithuania, China, South Korea, Russia, and New Zealand each had only one publication. These publications were based on different types of approaches; however, studies have discussed aspects related to the presence of CBs and the harm they cause in all these countries. One example is a study conducted by Barnes (2011), which highlighted the need for regulatory laws to consider CBs as a toxic waste in the United States, a country that has been a leading producer of tobacco and its derivatives for many years. In 2019, the production of tobacco and its by-products in the United States reached 212 260 tons/year, however, it currently ranks fifth worldwide (FAO 2022). This production can lead to the accumulation of CBs and its permanence in public areas, with the potential of becoming a chronic problem, as their compounds can persist for years in the environment (King et al. 2021; Dobaradaran et al. 2022).

Sometimes, the accumulation of CBs results from the personal, social, and structural habits of smokers. For example, in Iran, such habits influence the practice of irregular disposal, even though public clean-up is reported on a monthly basis (Akhbarizadeh et al. 2021*a*; Dobaradaran et al. 2022). According to Dehdari (2022), the reasons stated by smokers were the following: (1) laziness, (2) behaviour passed through generations, (3) fun, playing games with friends, and (4) lack of receptacles and environmental education. In this country, 43% of urban waste is composed of cigarette waste, with most of its elements being illegal, which makes its control even more difficult (Golestan et al. 2021; Jafari et al. 2021). On the other hand, information presented in reports based on data released by the Ministry of Industry, Mining, and Commerce of Iran indicate a significant increase in the manufacturing of national cigarettes. In 5 years, the number of companies in this industry increased by 228%, growing from 7 companies in 2015 to 23 in 2020 and increasing production from 29 billion cigarettes/year to 55 billion cigarettes/year (Financial Tribune 2022*a*; Italian Trade Agency 2022; Tehran Times 2022*b*). The lack of control over irregular disposal, significant amount of smuggling, and unbridled production may increasingly aggravate the socioenvironmental situation, becoming a complex motivator for the trivialisation of public health and environmental issues.

The magnitude of the production and consumption of cigarettes worldwide can influence the number of CBs discarded irregularly in the environment (Green et al. 2014). This fact can generate greater concern and monitoring of CB residues, which corresponds to an increase in the number of publications in those places. One example is in Italy where 24% of the population smoked (FAO 2022) while ranking fourth in the number of publications on environmental impacts caused by CBs. In other regions, such as Germany, CBs leached into rivers has also been analysed, where experiments using river water have highlighted a variety of compounds at different concentrations (Dobaradaran et al. 2019, 2021).

Brazil is the largest exporter and third largest producer of tobacco for cigarette production worldwide (FAO 2022), and coastal areas have been found to contain significant amounts of cigarette waste (Ribeiro et al. 2022). Scientists are concerned about the management of CBs and its collection on the Brazilian coast, as the vast majority of cleaning services are manual and inefficient for small items (Araújo and Costa 2019). The contributions provided by these countries can help promote the debate on the damage caused by CBs at a global level, given that these records can foster understanding with respect to the degree of the problem, as well as the local needs that can contribute to more connected and effective alternatives.

3.1. Lexicometric analyses

Based on the information provided, lexicometric and DHC analyses identified 116 abstracts with textual materials similar to each other, with the textual corpus divided into 472 text segments containing an efficiency of 87.92%. Therefore, the review of contamination from CBs resulted in the discovery of explicit and implicit aspects of the topic, with the formation of two groups with four classes, according to the DHC (Fig. 5). The grouping of classes 2 and 3 on the right in Fig. 5 consists of elements whose information is more qualitative with in situ sampling procedures, whereas the grouping of classes 1 and 4 depicted on the left has a predominance of quantitative data with ex situ analyses.

Class 1 vocabulary contextualized as “ecotoxicological studies” accounted for 29.2% of the analysed text segments, with content comprising information about the negative effects of CBs on the growth, reproduction, and survival of organisms. Class 2, described as “public policies”, was based on the management of the contaminant and possible actions that could help solve the problem, representing 25.5%. Class 3, entitled “contamination of public areas”, corresponding to 24.3%, highlighted the influence of human activity in open spaces. Finally, Class 4, called “physicochemical aspects”, represented 21% of the segments, with details of the elements present in CBs and the environmental damage they cause. These nominations were based on the vocabulary in

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Fig. 4. Worldwide distribution of studies published addressing environmental contamination from cigarette butts, between 2006 and 2022.

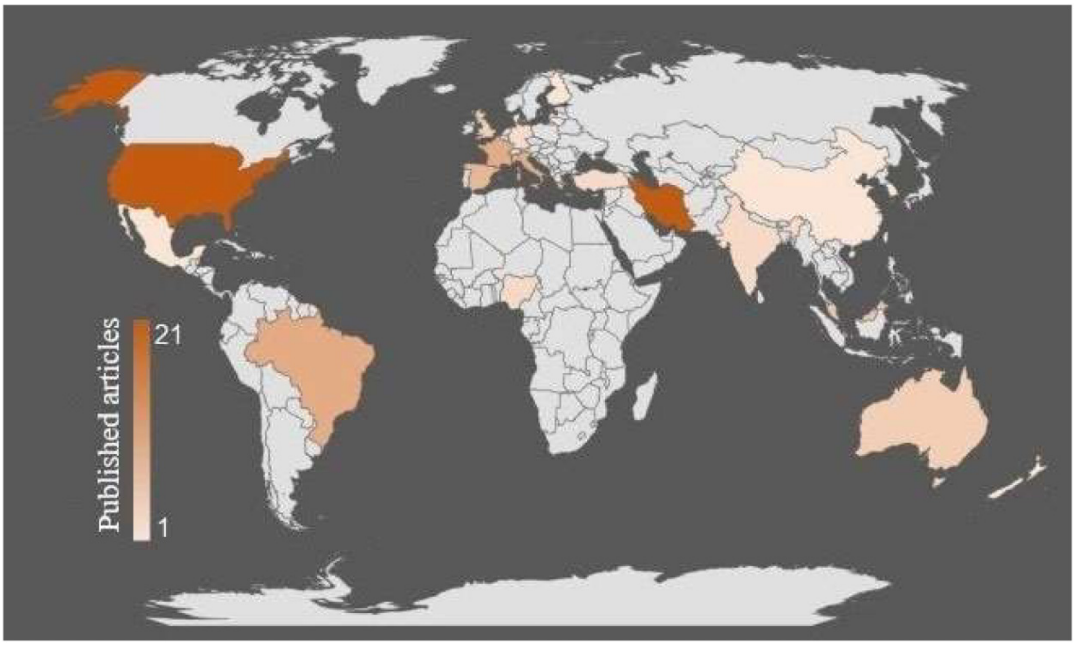
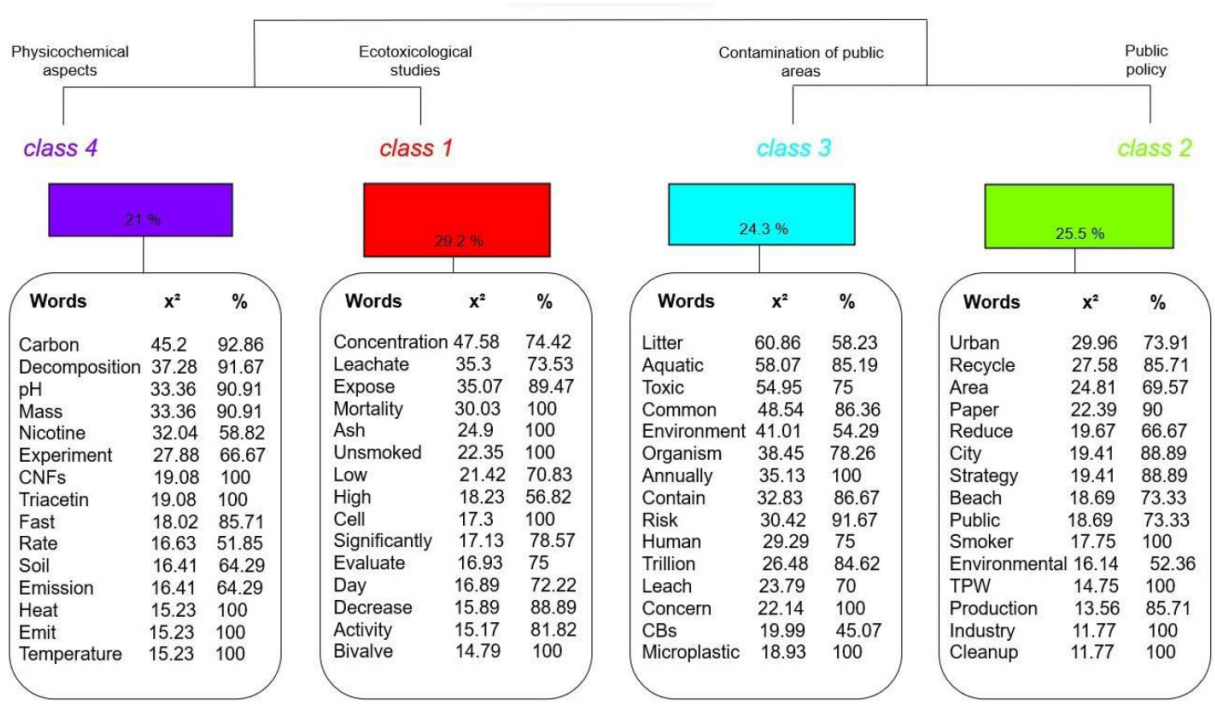


Fig. 5. Descending hierarchical classification of abstracts.



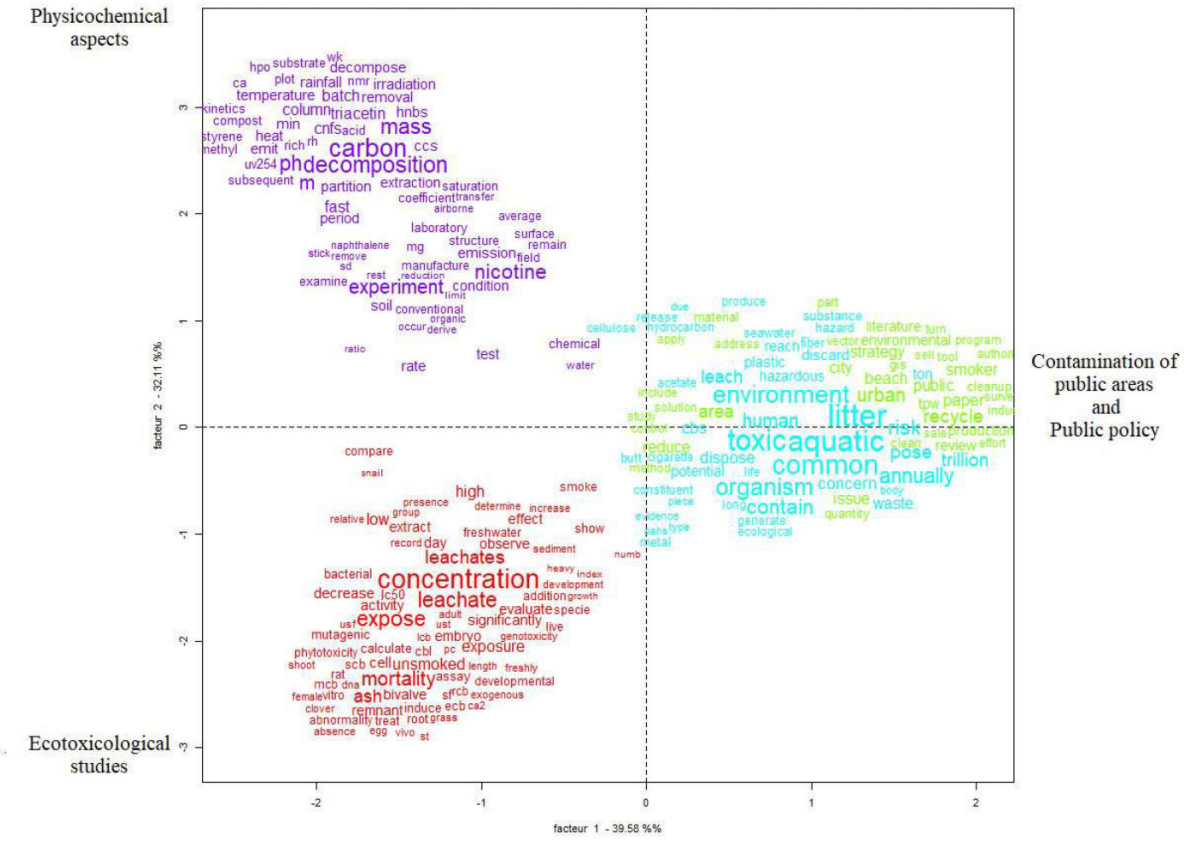
which the most representative words expressed the topics addressed in each class.

The ordering of classes could also be observed in the CFA where using a factorial plan and coordinates, it was possible to identify the connections and distances between the terms (Fig. 6).

For Factor 1, positioned on the horizontal axis, where 39.58% of the total variation was in elementary context units, with the occurrence of classes 1, 2, and 3. Factor 2, on the vertical axis, illustrates 32.11% of the text segments with the occurrence of classes 2, 3, and 4. This analysis allowed us to observe that, even though they corresponded to a single group in the CFA, the segments of classes 1 and 4 were distant, in terms with the same analytical profile, that is, with the prevalence of quantitative data and methods in artificial environments; however, even with these similarities they still

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Fig. 6. Correspondence factor analysis of the selected studies.



had opposite or very specific concepts, which may explain the positioning of these classes in the plan. However, the same fact was not observed in the other groups, since classes 2 and 3 represented related concepts and were also determined by the same scientific method, where most of the data were qualitative and the studies were conducted in habitats or natural environments. Having similar methodologies and intertwined subjects may explain the closeness of classes 2 and 3 in the plan.

According to the data obtained, it can be considered that, in addition to the variation between publications, there was also a co-occurrence of interconnected information throughout the studies, highlighting the importance of combining different methods and concepts in favour of expanding information on the subject.

3.1.1. Ecotoxicological studies

Class 1, corresponding to ecotoxicology studies, had the largest representation among all the classes. The vocabulary presented in this cluster highlights information about the methods used and the toxic effects of CBs on biodiversity. The word "concentration" ($x^2 = 47.28$) exhibited the highest significance value, whereas "mortality", "ash", "unsmoked", "cell", and "bivalve" were the most frequent (100%). The reviewed studies used organisms that were adapted to different environments. In addition, they used different methods based on the types, concentrations, and exposure times of the CB leachate (Table 2). This review compiled data on CBs that emphasized its high potential for environmental contamination, regardless of the degree of degradation of the material or level of dilution of the leachate. Contamination from CB leachate occurs through the displacement of contaminants into the environment through the action of water. Toxic compounds present in filters and tobacco residues can release contaminants that cause damage to organisms (Micevska et al. 2006).

All publications addressing class 1 topics indicated some sublethal effects on fish, amphibians, plants, and invertebrates (of which bivalves were the most representative). Mortality caused by exposure to CB leachate has been recorded in studies conducted by Lawal and Ologundudu (2013), Booth et al. (2015), and Green et al. (2020). The bioaccumulation of metals (Cr, Ni, Pb, Mn, and Zn) from CBs is associated with changes in the behaviour of adult animals (Montalvão et al. 2019*a*). The same authors also assessed the mutagenic potential by observing mutations in mollusc cells caused by exposure to CB contaminants (Montalvão et al. 2019*b*), and reinforced that the severity of CBs disposal went beyond their physical representation. In addition, the selected studies from this review provided data related to the categories of CBs, such as traditional, biodegradable, menthol, smoked or nonsmoked electronic cigarettes, ability to cause harm to organisms, and dynamics of their populations. These studies have highlighted that similar alternative materials can also cause harm.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Table 2. Main information extracted from ecotoxicology studies addressing lethal and sublethal effects of cigarette butts on biota.

Reference	Test organism	Method	Environmental impact
Green et al. (2020)	*Dreissena polymorpha*, *Polycelis nigra*, *Planorbis planorbis*, and *Bithynia tentaculata*	Two types of butts were used: (1) cellulose filters and (2) cellulose acetate filters, all smoked, with three different concentrations: 0.2, 1, and 5 CB/L.	Leachate derived from 5 CB/L of any type of filter caused 60%–100% mortality in all species within 5 days. The leachate derived from 1 CB/L of any type exhibited less activity in adults compared to none or 0.2 CB/L.
Green et al. (2019)	*Lolium perenne* L. and *Trifolium repens* L.	The experiment w as conducted with the exposure of traditional and menthol CBs, at three different levels: nonsmoked, smoked (without tobacco residue), and smoked with remaining tobacco, for 21 days.	For *Lolium perene*, there was a decrease of approximately 10%, 13%, and 18% in germination, shoot growth, and chlorophyll *b*, respectively, for any type of CBs. Whereas for *Trifolium repens*, there was a reduction of approximately 27%, 28%, and 57% in germination, shoot length, and biomass, respectively.
Montalvão et al. (2019*a*)	*Anodontitis trapesialis*	The study used 66 adult specimens, with smoked 20 CB/L and dilutions of 1.375 and 13.75 CBs for a 14 day exposure.	Sublethal effect was observed with atypical digging behaviours. High concentrations of metals were associated with behavioural changes observed in animals exposed to the two dilutions.
Montalvão et al. (2019*b*)	*Anodontitis trapesialis*	The study used 66 adult individuals, with smoked 20 CB/L and dilutions of 1.375 and 13.75 CBs for a 14 day exposure, with analysis in type I and II hyalinocytes.	Mutagenic potential of CB leachate as nuclear abnormalities were higher in type I hyalinocytes from bivalve individuals exposed to the contaminant than in those of individuals from the control group.
Osuala et al. (2017)	*Oreochromis niloticus*	Organisms were exposed to two different brands of CBs, using smoked and nonsmoked specimens, with a concentration of 1–5 CB/L.	After 96 h, the study indicated a difference in toxicity between the brands, being classified according to their LC_{50} value: (I) brand 1 smoked butts were considered the most toxic (1.346 CB/L), followed by (II) brand 2 smoked butts (2.271 CB/L), (III) brand 2 nonsmoked butts (7.313 CB/L), and (IV) brand 1 nonsmoked butts (5.559 CB/L).
Parker and Rayburn (2017)	*Xenopus laevis*	Embryos were exposed to butt leachate at concentrations of 0–4 CB/L for traditional and menthol CBs, and concentrations of 0–10 CB/L for e-cigarette butts, for a period of 96 h.	The study reported that traditional CB leachate was the most toxic compound, whereas the menthol CB leachate had the greatest teratogenicity. The highest value for LC_{50} was reported for traditional and menthol CB leachate, with LC_{50} of approximately 1 CB/L. Significant differences in malformation related to leachate from traditional CBs were also observed.
Lee and Lee (2015)	*Oryzias latipes*	The embryos were exposed with the leachates having the respective concentrations: (1) nonsmoked tobacco, (2) smoked tobacco, (3) nonsmoked filter, (4) smoked filter; (1) and (2) having 16 segments with 1 cm; whereas (3) and (4) had 16 cigarette filters. Each group was soaked in 800 mL of solution.	From 1–3 days after fertilization, the embryos were observed, and toxicity of the leachates was found with the following classifications: (2) > (1) > (4) > (3). Low concentrations of leachates in (2), (1), and (4) increased heart rate, accelerated development, and changed behaviour, whereas high concentrations reduced heart rate, suppressed development, and increased mortality.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Table 2. (*concluded*).

Reference	Test organism	Method	Environmental impact
Booth et al. (2015)	*Austrocochlea porcata*, *Nerita atramentosa*, and *Bembicium nanum*	CB leachate with the following concentrations: 0.25, 0.50, 1.25, and 5 CB/L.	After 8 days of experiment, 100% mortality was recorded for the three species assessed in the highest concentration. Furthermore, *Austrocochlea porcata* exhibited greater mortality than the other two species at concentrations of 0.50 and 1.25 CB/L.
Lawal and Ologundudu (2013)	*Hymenochirus curtipes* and *Clarias gariepinus*	Smoked and nonsmoked cigarette filters with concentrations of 0.625–10 g/L, and exposure time of 96 h.	Mortality rates increased in both organisms with a significant difference only after 72 and 96 h of exposure. The smoked CB leachate was approximately 2.5 times more toxic than the nonsmoked CB leachate. The lethality of *Hymenochirus curtipes* and *Clarias gariepinus* related to smoked CB leachate was six times greater than that caused by the nonsmoked filter leachate after 24 h exposure with an LC_{50} of 8.73 g/L.
Slaughter et al. (2011)	*Atherinops affinis* and *Pimephales promelas*	Leachates from smoked and nonsmoked CBs were used at a concentration of 8 CB/L, for 4 days.	Leachate from smoked CBs was toxic to both species. The smoked CB leachate (no tobacco) was less toxic, with LC_{50} values of 1.8 and 4.3 CB/L for *Atherinops affinis* and *Pimephales promelas*, respectively. Nonsmoked CBs (nontobacco) and cigarette filters were also considered toxic, with LC_{50} values of 5.1 and 13.5 CB/L, respectively, for both fish species.

Note: Abbreviation: CB, cigarette butt.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Other studies have emphasized that compounds found associated with CBs, such as pesticides, herbicides, and insecticides, could release metals into tobacco leaves, further accentuating environmental degradation (Zoffoli et al. 2013). Furthermore, cigarette ash can cause DNA damage and, consequently, a decline in biological diversity (Mansouri et al. 2020). Given the information obtained through ecotoxicological studies, it is evident that CBs and its irregular disposal have high potential to contaminate the environment, which can lead to the mortality of various organisms, in addition to the breakdown of biological diversity in different environments.

3.1.2. Public policies

With the second largest representation, class 2 (public policies) had the word "urban" ($\chi^2 = 29.96$), with the highest significance. The words "tobacco product waste", "industry", and "clean-up" had the highest frequency (100). The studies that addressed this topic evaluated precaution to avoid environmental impacts and health risks from these wastes (Table 3). The starting point with respect to the theme and care for human health and the environment can be observed in publications showing data in urban and marine areas, which were generated via cleaning actions. This reveals the need for the involvement of several sectors, as indicated in studies conducted by Marah and Novotny (2011), Kataržytė et al. (2020), and Valiente et al. (2020). Policies for reducing the environmental impact caused by CBs, also observed in the present review, suggest a set of behaviours as an option to reduce the problem, such as those mentioned by Novotny et al. (2009).

(I) Proper labelling of products, denoting their composition and tips for regular disposal. This measure could encourage consumers to be more careful regarding this waste.
(II) Implementation of "deposit/recycling laws", including a "butt deposit", which requires consumers to pay a fee when purchasing specific items. These amounts are refunded when the packs are returned to the same location. This measure could reduce irregular disposal, and portable devices to store CBs could be sold at collection sites.
(III) Actualization of a consumer-financed recycling tax, encouraging shared responsibility. This measure could reduce consumption, in addition to reducing environmental damage, by using the funds obtained from those taxes in projects aimed at understanding the disposal of CBs as well as to finance cleaning campaigns.

Table 3. Other studies that address public policy issues regarding the improper disposal of CBs.

Reference	Approaches in the studies
Dehdari (2022)	From 27 interviews we observed smokers' experiences with CBs disposal in public areas, where categories such as personal, social, and structural factors were the main information discussed by respondents for the behaviour.
Smith and Novotny (2011)	Research conducted at the Legacy Tobacco Documents Library, identified that tobacco companies have studied smokers' beliefs, behaviours, and attitudes about smokers' ideas for new products. However, these proposals made to reduce contamination from CBs are not as effective.
Novotny et al. (2009)	Public policy approaches addressed in observed studies brought two benefits to health and the environment. First: with the increased cost could reduce smoking. Second: government, voluntary groups, and outside agencies bear most of the costs of CBs clean-up.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

(IV) Actions taken to associate tobacco companies/industries with impacts related to product sales. Cigarette-producing companies would start paying through litigation to take responsibility for the environmental impacts associated with cigarette sales. Currently, most of these actions aim to defray the impact on health; however, it would be of interest to simultaneously charge taxes aimed at the environment.

(V) Fines and charges for violations of the smoking ban in inappropriate places or illegal disposal, focusing on individuals, companies, states, or municipalities.

(VI) Set of regulations for tobacco production aimed at better standardisation. This would yield greater control of the compounds and materials present in cigarettes.

(VII) Bans on the sale of cigarettes with filters made of cellulose acetate, requiring more research on the subject, since other types of filters such as biodegradable ones also raise doubts as to whether they could be used as an alternative material in place of traditional CBs.

(VIII) Environmental and social education campaigns addressing responsibility for the disposal of CBs in suitable places financed by the tobacco industry.

This type of understanding can support the creation of public policies related to the disposal of CBs and the environment and can positively influence both environmental and social perspectives. However, it is worth emphasising that different environments, urban areas, and regions tend to have local needs, where these actions would also be shaped according to each locality.

Likewise, there is a need to determine several implementations (or even simultaneous implementations) for greater efficiency in reducing irregular CBs disposal. An example was reported by Kataržytė et al. (2020), who assessed the installation of butt containers on beaches in Germany and Lithuania. These authors observed that the use of these devices alone was not as effective in reducing CBs in the areas, highlighting the need to strengthen measures and/or implement other measures for a more efficient reduction of this contaminant.

A study conducted with members of the Framework Convention Alliance for Tobacco Control assessed knowledge, beliefs, awareness, and perceptions of CBs based on responses to questionnaires (Granados et al. 2019). With 65 individuals from 37 countries interviewed, the survey emphasized that the opinions of members of these types of organisations exerted greater influence on government agencies, decision-making, and contributions to the control of CBs waste.

The sum of activities and studies addressing environmental contamination by CBs should not be isolated or with very specific approaches and issues. On the contrary, they should be interconnected in several areas to maximize knowledge about this type of impact and consequently facilitate actions aimed at reducing contamination. Even in the presence of gaps, this objective was predominant in the studies that were evaluated.

3.1.3. Contamination of public areas

Class 3 (contamination of public areas) showed high significance for the word "litter" ($x^2 = 60.86$), whereas the words "annually", "concern", and "microplastic" exhibited the highest frequency (100%). Class 3 concepts were based on the social and environmental implications of the irregular disposal of CBs in public environments. Despite the values obtained in this class, the vocabulary related to "contamination in public areas" was characterized by documental bases and reviews as a large part of the structure of the texts (Table 4).

Spaces with large circulation of people, such as streets, squares, parks, gardens, commercial areas, and beaches, tend to have a large amount of CBs waste. According to Rahman et al. (2020), approximately 75% of smokers irregularly dispose CBs. High amounts of CBs have been recorded in studies from different regions, including approximately 36 757 CBs counted in the city of Qazvin, Iran (Torkashvand et al. 2021) and 4174 CBs along eight Brazilian beaches (Araújo and Costa 2021). In addition to being universal, it is worth noting that this problem is also associated with the movement of individuals. Another factor influencing the disposal of CBs in the environment is related to tourism, with a high abundance of this material observed mainly during the summer season, comprising 46% of the objects observed on the coast of Spain (Martinez-Ribes et al. 2007) and 51.5% of items registered in Brazil during the same season (Lima et al. 2021). The accumulation of this contaminant in such high numbers can have numerous environmental effects.

Studies conducted in public spaces or analyses performed with materials collected in these regions were based on the

Table 4. Main studies on CBs contamination in public areas, with information in natural environments from other comprehensive systematic reviews.

Reference	Type of contamination
Venugopal et al. (2021)	It was found that one-third of the 98 chemical compounds present in cigarette butts collected from marine environments were highly toxic, with 10% being toxic to aquatic organisms.
Shen et al. (2021)	The ecological effects of microplastic fibers and pollutants released by CBs in natural environments have been observed.
Dobaradaran et al. (2021)	It analysed damage caused by CBs in aquatic environments, informing the need for regulation on the compounds present in the material.
Santos-Echeandía et al. (2021)	Accumulation of the metals present in tobacco products was verified under real-world ambient conditions, with copper being the compound with the highest percentage of desorption.

Table 5. Main information extracted from studies on physicochemical aspects of the CBs.

Reference	Evaluation type	Results
Gill et al. (2018)	Effects of CBs on terrestrial organism	In soil, the CBs showed a decline in toxicity directly proportional to the aging of the material.
Selmar et al. (2018)	CBs nicotine adsorption in plants	Plant and soil contamination was observed even with the presence of only 1 CB/m^2.
Bonanomi et al. (2015)	CB degradation	A reduction of almost 40% of the initial CBs mass after 2 years was observed, possessing a low degradation rate.
Green et al. (2014)	Effects of CBs on urban water resources	According to these results observed in the study there was a rapid release of nicotine in rain simulation experiments, where one CB could contaminate 1000 L of water.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

premise of studying the damage caused by CB fragments in areas where the product was discarded. According to a study conducted by Venugopal et al. (2021), 109 CBs collected in areas intended for smokers, restaurants, and commercial enterprises in Pennsylvania (United States) identified 98 chemical products, of which 25 were determined to be harmful or potentially harmful according to the Food and Drug Administration list. Furthermore, Santos-Echeandía et al. (2021) evaluated smoked CBs in two areas (one on a beach and the other in a marina) in France, where they assessed that copper recorded the highest percentage of release (L = 91%), warning about the cumulative potential of CBs and metal adsorption from the environment. Dobaradaran et al. (2017) and Soleimani et al. (2021) monitored the levels of compounds present in CBs in nine coastal areas of Bushehr, Persian Gulf and identified Cd, Fe, As, Ni, Cu, Zn, and Mn at considerable levels that could generate environmental degradation. These assessments are of great importance because they were conducted in public spaces where more realistic information can be obtained, as natural environments are not areas with conditions controlled by humans. They can also generate contributions from studies that analyse the consequences of CBs, in addition to assisting in projections of future damage.

3.1.4. Physicochemical aspects

Among the groups, class 4 (physicochemical aspects) had the smallest representation (21%) and the greatest significance for the word "carbon" (x^2 = 45.2). The words "cellulose nanofibers", "triacetin", "heat", "emit", and "temperature" were observed with 100% frequency. The content analysed in this class made it possible to obtain information that indicated the structures, physicochemical modifications, and interactions of the properties of CBs in the environment, as shown in the studies in Table 5.

The adsorbed potentially toxic elements can become bioavailable through leaching from CBs, reaching water resources (Akhbarizadeh et al. 2021*b*). A single cigarette may contain PAHs (polycyclic aromatic hydrocarbon), benzene, toluene, ethylbenzene, *o*-xylene, *p*-xylene, and heavy metals, which can be harmful to humans and aquatic organisms (Dobaradaran et al. 2021). Naphthalene and fluorene have high concentrations among the PAHs analysed by Dobaradaran et al. (2020).

After the production, distribution, and delivery of cigarettes to points of sale, the product is ready for smoking. After the cigarette is smoked, its composition is converted into two categories: (1) nonvolatile or semivolatile chemicals and (2) volatile chemicals (Kurmus and Mohajerani 2020). When extinguished, cigarettes can still spread compounds that can be influenced by other factors. For example, temperature and relative humidity can affect chemical emissions from CBs, resulting in the faster release of nicotine, triacetin, and other chemicals (Poppendieck et al. 2020). Some CBs disposal sites may contain greater amounts of contaminants, owing to the environmental conditions.

According to Belzagui et al. (2021), each CB has 15 000 cellulose acetate threads and can release approximately 100 threads per day. Most of these threads are smaller than 0.2 mm and, consequently, are an important source of

microplastics that directly affect aquatic and trophic chains. This degradation of CBs occurs very slowly because it has a reduced number of nutrients, mainly nitrogen. Nitrogen has a regulatory role during this process, as CB has a reduction rate of only 37.8% over 2 years (Bonanomi et al. 2015). In addition to traditional filters (composed of acetate), there are other types of biodegradable filters (composed of cellulose) that are considered less harmful to the environment. However, this only occurs when CB is under controlled conditions and not smoked, because after smoking, tar and other chemical products are adsorbed and significantly slow down decomposition rates (Joly and Coulis 2018). In an attempt to reduce health damage, cigarette filters were adapted to retain substances present in tobacco. Thus, studies that assess the dynamics and implications of CBs, by considering physical and chemical factors, can also make significant contributions to the subject.

4. Conclusion

Contamination by CBs is a global problem that significantly affects the environment and human health. Although the theme addresses different areas and studies, in general, there is an absence of improvement and proposals to combat the irregular disposal of CBs in the environment. Our systematic literature review highlights gaps in all four classes: (1) ecotoxicological studies, (2) public policy, (3) contamination of public areas, and (4) physicochemical aspects. In addition to the small number of studies on ecotoxicology with a limitation in the diversity of bioassays, there was also a lack of studies addressing large-scale public policy application and comparisons with similar issues that could help promote collective action. We also found that in studies with records of contaminants in public areas, no warnings were given regarding the problems caused by CBs. Furthermore, there is a lack of studies addressing the degradation of CBs in natural environments and the influence of its compounds under environmental conditions (e.g., weathering). This systematic review underscored the worldwide dispersion of CBs and their power of accumulation, degradation, and toxicity, which go beyond human health issues to directly impact the environment.

Acknowledgements

We thank everyone who contributed to the development of the study and the Oceanography and Water Pollution Laboratory of the Federal Rural University of Pernambuco for their structural support.

Article information

History dates

Received: 23 July 2022
Accepted: 23 November 2022
Version of record online: 10 February 2023

Copyright

© 2023 The Author(s). Permission for reuse (free in most cases) can be obtained from copyright.com.

Data availability

The data generated or analysed in this study are provided in full within the published article.

Author information

Author ORCIDs

Jacqueline Santos Silva-Cavalcanti https://orcid.org/0000-0001-6859-4325

Author contributions

Conceptualization: NFS, JSS-C
Data curation: NFS, MCBA
Formal analysis: NFS, JSS-C
Investigation: NFS
Methodology: NFS, JSS-C
Supervision: MCBA
Validation: MCBA, JSS-C
Writing – original draft: NFS
Writing – review & editing: MCBA, JSS-C

Competing interests

The authors declare there are no competing interests.

Funding information

The authors declare no specific funding for this work.

References

Akhbarizadeh, R., Dobaradaran, S., Nabipour, I., Tangestani, M., Abedi, D., Javanfekr, F., et al. 2021b. Abandoned Covid-19 personal protective equipment along the Bushehr shores, the Persian Gulf: an emerging source of secondary microplastics in coastlines. Mar. Pollut. Bull. **168**: 112386. doi:10.1016/j.marpolbul.2021.112386. PMID: 33901902.

Akhbarizadeh, R., Dobaradaran, S., Parhizgar, G., Schmidt, T.C., and Mallaki, R. 2021a. Potentially toxic elements leachates from cigarette butts into different types of water: a threat for aquatic environments and ecosystems? Environ. Res. **202**: 111706. doi:10.1016/j.envres.2021.111706. PMID: 34284015.

Araújo, M.C.B., and Costa, M.F. 2007. Visual diagnosis of solid waste contamination of a tourist beach: Pernambuco, Brazil. Waste Manag. **27**: 833–839. doi:10.1016/j.wasman.2006.04.018. PMID: 16842985.

Araújo, M.C.B., and Costa, M.F. 2019. A critical review of the issue of cigarette butt pollution in coastal environments. Environ. Res. **172**: 137–149. doi:10.1016/j.envres.2019.02.005. PMID: 30782533.

Araújo, M.C.B., and Costa, M.F. 2021. Cigarette butts in beach litter: snapshot of a summer holiday. Mar. Pollut. Bull. **172**: 112858. doi:10.1016/j.marpolbul.2021.112858.

Asensio-Montesinos, F., Ramírez, M.O., González-Leal, J.M., Carrizo, D., and Anfuso, G. 2020. Characterization of plastic beach litter by Raman spectroscopy in South-western Spain. Sci. Total Environ. **744**: 140890.

Barnes, R.L. 2011. Regulating the disposal of cigarette butts as toxic hazardous waste. Tob. Control, **20**: 45–48. doi:10.1136/tc.2010.041301.

Becherucci, M.E., Rosenthal, A.F., and Pon, J.P.S. 2017. Marine debris in beaches of the Southwestern Atlantic: an assessment of their abundance and mass at different spatial scales in northern coastal Argentina. Mar. Pollut. Bull. **119**: 299–306. doi:10.1016/j.marpolbul.2017.04.030. PMID: 28434664.

Belzagui, F., Buscio, V., Gutiérrez-Bouzán, C., and Vilaseca, M. 2021. Cigarette butts as a microfiber source with a microplastic level of concern. Sci. Total Environ. **762**: 144165. doi:10.1016/j.scitotenv.2020.144165.

Benzécri, J.P. 1973. L'analyse des données(vol. **2**, pp. l). Dunod, Paris.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.

Bonanomi, G., Incerti, G., Cesarano, G., Gaglione, S.A., and Lanzotti, V. 2015. Cigarette butt decomposition and associated chemical changes assessed by 13C CPMAS NMR. PLoS ONE, **10**: e0117393. doi:10.1371/journal.pone.0117393.

Booth, D.J., Gribben, P., and Parkinson, K. 2015. Impact of cigarette butt leachate on tidepool snails. Mar. Pollut. Bull. **95**: 362–364. doi:10.1016/j.marpolbul.2015.04.004. PMID: 25913792.

De-la-Torre, G.E., Dioses-Salinas, D.C., Dobaradaran, S., Spitz, J., Keshtkar, M., Akhbarizadeh, R., et al. 2022a. Physical and chemical degradation of littered personal protective equipment (PPE) under simulated environmental conditions. Mar. Pollut. Bull. **178**: 113587. doi:10.1016/j.marpolbul.2022.113587. PMID: 35397345.

De-la-Torre, G.E., Dioses-Salinas, D.C., Pizarro-Ortega, C.I, Severini, M.D.F., López, A.D.F., Mansilla, R., et al. 2022b. Binational survey of personal protective equipment (PPE) pollution driven by the COVID-19 pandemic in coastal environments: abundance, distribution, and analytical characterization. J. Hazard. Mater. **426**: 128070. doi:10.1016/j.jhazmat.2021.128070. PMID: 34922133.

Dehdari, T. 2022. A qualitative exploration of Iranian smokers' experiences in terms of cigarette butt littering behaviour. Int. J. Environ. Health Res. **32**: 417–425. doi:10.1080/09603123.2020.1769040. PMID: 32436392.

Dobaradaran, S., Mutke, X.A., Schmidt, T.C., Swiderski, P., De-la-Torre, G.E., and Jochmann, M.A. 2022. Aromatic amines contents of cigarette butts: fresh and aged cigarette butts vs unsmoked cigarette. Chemosphere, **301**: 134735. doi:10.1016/j.chemosphere.2022.134735. PMID: 35489462.

Dobaradaran, S., Nabipour, I., Saeedi, R., Ostovar, A., Khorsand, M., Khajeahmadi, N., et al. 2017. Association of metals (Cd, Fe, As, Ni, Cu, Zn and Mn) with cigarette butts in northern part of the Persian Gulf. Tob. Control, **26**: 461–463. doi:10.1136/tobaccocontrol-2016-052931. PMID: 27384342.

Dobaradaran, S., Schmidt, T.C., Kaziur-Cegla, W., and Jochmann, M.A. 2021. BTEX compounds leachates from cigarette butts into water environment: a primary study. Environ. Pollut. **269**, 116185. doi:10.1016/j.envpol.2020.116185.

Dobaradaran, S., Schmidt, T.C., Lorenzo-Parodi, N., Jochmann, M.A., Nabipour, I., Raeisi, A., et al. 2019. Cigarette butts: an overlooked source of PAHs in the environment? Environ. Pollut. **249**: 932–939. doi:10.1016/j.envpol.2019.03.097. PMID: 30965545.

Dobaradaran, S., Schmidt, T.C., Lorenzo-Parodi, N., Kaziur-Cegla, W., Jochmann, M.A., Nabipour, I., et al. 2020. Polycyclic aromatic hydrocarbons (PAHs) leachates from cigarette butts into water. Environ. Pollut. **259**: 113916. doi:10.1016/j.envpol.2020.113916. PMID: 32023793.

Dobaradaran, S., Schmidt, T.C., Nabipour, I., Khajeahmadi, N., Tajbakhsh, S., Saeedi, R., et al. 2018a. Characterization of plastic debris and association of metals with microplastics in coastline sediment along the Persian Gulf. Waste Manag. **78**: 649–658. doi:10.1016/j.wasman.2018.06.037. PMID: 32559956.

Dobaradaran, S., Schmidt, T.C., Nabipour, I., Ostovar, A., Raeisi, A., Saeedi, R., et al. 2018b. Cigarette butts abundance and association of mercury and lead along the Persian Gulf beach: an initial investigation. Environ. Sci. Pollut. Res. **25**(6): 5465–5473. doi:10.1007/s11356-017-0676-9. PMID: 29214478.

El Hadri, H., Lisa, J.M., Gigault, J., Reynaud, S., and Grassl, B. 2021. Fate of nanoplastics in the environment: implication of the cigarette butts. Environ. Pollut. **268**: 115170. doi:10.1016/j.envpol.2020.115170.

Fauziah, S.H., Rizman-Idid, M., Cheah, W., Loh, K.H., Sharma, S. Noor-Maiza, M.R., et al. 2021. Marine debris in Malaysia: a review on the pollution intensity and mitigating measures. Mar. Pollut. Bull. **167**: 112258. doi:10.1016/j.marpolbul.2021.112258.

Financial Tribune. 2022a. Cigarette prices to rise by 10–30%. Available from https://financialtribune.com/articles/domestic-economy/108287/cigarette-prices-to-rise-by-10-30 [accessed 25 June 2022].

Financial Tribune. 2022b. Number of Iran's tobacco companies on the rise. Available from https://financialtribune.com/articles/domestic-economy/105726/number-of-iran-s-tobacco-companies-on-the-rise [accessed 25 June 2022].

Food and Agriculture Organization (FAO). 2022. Available from http://www.fao.org/faostat/en/#data/QCL/visualize [accessed 25 June 2022].

Gill, H., Rogers, K., Rehman, B., Moynihan, J., and Bergey, E.A. 2018. Cigarette butts may have low toxicity to soil-dwelling invertebrates: evidence from a land snail. Sci. Total Environ. **628**: 556–561. doi:10.1016/j.scitotenv.2018.02.080. PMID: 29454197.

Golestan, Y.P., Ebrahimi Kalan, M., Ben Taleb, Z., Ward, K.D., Fazlzadeh, M., Bahelah, R., et al. 2021. The effect of price on cigarette consumption, distribution, and sale in Tehran: a qualitative study. BMC Public Health, **21**: 1–9. doi:10.1186/s12889-021-11733-5. PMID: 33388037.

Granados, P.S., Fulton, L., Nunez Patlan, E., Terzyk, M., and Novotny, T.E. 2019. Global health perspectives on cigarette butts and the environment. Int. J. Environ. Res. Public Health, **16**: 1858. doi:10.3390/ijerph16101858. PMID: 31130709.

Green, A.L.R., Putschew, A., and Nehls, T. 2014. Littered cigarette butts as a source of nicotine in urban waters. J. Hydrol. **519**: 3466–3474. doi:10.1016/j.jhydrol.2014.05.046.

Green, D.S., Boots, B., Carvalho, J.D.S., and Starkey, T. 2019. Cigarette butts have adverse effects on initial growth of perennial ryegrass (Gramineae: *Lolium perenne* L.) and white clover (Leguminosae: *Trifolium repens* L.). Ecotoxicol. Environ. Saf. **182**:109418. doi:10.1016/j.ecoenv.2019.109418.

Green, D.S., Kregting, L., and Boots, B. 2020. Smoked cigarette butt leachate impacts survival and behaviour of freshwater invertebrates. Environ. Pollut. **266**: 115286. doi:10.1016/j.envpol.2020.115286.

Grillo, A.C., and Mello, T.J. 2021. Marine debris in the Fernando de Noronha Archipelago, a remote oceanic marine protected area in tropical SW Atlantic. Mar. Pollut. Bull. **164**: 112021. doi:10.1016/j.marpolbul.2021.112021. PMID: 33515831.

Italian Trade Agency. 2022. Number of Iran's tobacco companies on the rise. Available from https://www.ice.it/it/news/notizie-dal-mondo/161254 [accessed 25 June 2022].

Jafari, A.J., Latifi, P., Kazemi, Z., Kazemi, Z., Morovati, M., Farzadkia, M., and Torkashvand, J. 2021. Development a new index for littered waste assessment in different environments: a study on coastal and urban areas of northern Iran (Caspian Sea). Mar. Pollut. Bull. **171**: 112684. doi:10.1016/j.marpolbul.2021.112684.

Joly, F.X., and Coulis, M. 2018. Comparison of cellulose vs. plastic cigarette filter decomposition under distinct disposal environments. Waste Manag., **72**: 349–353. doi:10.1016/j.wasman.2017.11.023. PMID: 29153904.

Kadir, A.A., and Sarani, N.A. 2015. Cigarette butts pollution and environmental impact–a review. Appl. Mech. Mater. **773–774**: 1106–1110. doi:10.4028/www.scientific.net/AMM.773-774.1106. Trans Tech Publications Ltd.

Kataržytė, M., Balčiūnas, A., Haseler, M., Sabaliauskaitė, V., Lauciūtė, L., Stepanova, K., et al. 2020. Cigarette butts on Baltic Sea beaches: monitoring, pollution and mitigation measures. Mar. Pollut. Bull. **156**: 111248. doi:10.1016/j.marpolbul.2020.111248.

King, I.C., Lorenzi, V., Blasius, M.E., and Gossett, R. 2021. Leachates from cigarette butts can persist in marine sediment. Water Air Soil Pollut. **232**: 1–13. doi:10.1007/s11270-021-04999-3.

Kurmus, H., and Mohajerani, A. 2020. The toxicity and valorization options of cigarette butts. Waste Manag. **104**: 104–118. doi:10.1016/j.wasman.2020.01.011.

Lawal, M.S., and Ologundudu, S.O. 2013. Toxicity of cigarette filter leachates on *Hymenochirus curtipes* and *Clarias gariepinus* in Nigeria. J. Environ. Ext. **11**: 7–14.

Lee, W., and Lee, C.C. 2015. Developmental toxicity of cigarette butts–an underdeveloped issue. Ecotoxicol. Environ. Saf. **113**: 362–368. doi:10.1016/j.ecoenv.2014.12.018.

Lima, C.F., Santos, M.A.P., Choueri, R.B., Moreira, L.B., and Castro, Í.B. 2021. Occurrence, characterization, partition, and toxicity of cigarette butts in a highly urbanized coastal area. Waste Manag. **131**: 10–19. doi:10.1016/j.wasman.2021.05.029.

Lozano-Rivas, W.A., Salinas, A., and Rommel Bonilla, C. 2020. Estimation of potential pollution of cigarette butts littered in nightlife areas in Bogota D.C. upon its river. Int. J. Res. Stud. Sci. Eng. Technol. **7**: 22–28.

Mansouri, N., Etebari, M., Ebrahimi, A., Ebrahimpour, K., Rahimi, B., and Hassanzadeh, A. 2020. Genotoxicity and phytotoxicity comparison of cigarette butt with cigarette ash. Environ. Sci. Pollut. Res. **27**: 40383–40391. doi:10.1007/s11356-020-10080-z.

Marah, M., and Novotny, T.E. 2011. Geographic patterns of cigarette butt waste in the urban environment. Tob. Control, **20**: 42–44. doi:10.1136/tc.2010.042424.

Marin, C.B., Niero, H., Zinnke, I., Pellizzetti, M.A., Santos, P.H., Rudolf, A.C., et al. 2019. Marine debris and pollution indexes on the beaches

of Santa Catarina State, Brazil. Reg. Stud. Mar. Sci. **31**: 100771. doi:10.1016/j.rsma.2019.100771.

Martinez-Ribes, L., Basterretxea, G., Palmer, M., and Tintoré, J. 2007. Origin and abundance of beach debris in the Balearic Islands. Sci. Mar. **71**: 305–314.

Micevska, T., Warne, M.S.J., Pablo, F., and Patra, R. 2006. Variation in, and causes of, toxicity of cigarette butts to a cladoceran and microtox. Arch. Environ. Contam. Toxicol. **50**: 205–212. doi:10.1007/s00244-004-0132-y.

Montalvão, M.F., Chagas, T.Q., da Silva, T.G.A, Mesak, C., Costa, A.P.A., Gomes, A.R., et al. 2019b. How leachates from wasted cigarette butts influence aquatic life? A case study on freshwater mussel *Anodontites trapesiali*. Sci. Total Environ. **689**: 381–389. doi:10.1016/j.scitotenv.2019.06.385.

Montalvão, M.F., Chagas, T.Q., Silva, T.G.A., Mesak, C., Araújo, A.P.C., Gomes, A.R., et al. 2019a. Cigarette butt leachate as a risk factor to the health of freshwater bivalve. Chemosphere, **234**: 379–387. doi:10.1016/j.chemosphere.2019.06.100.

Novotny, T.E., and Slaughter, E. 2014. Tobacco product waste: an environmental approach to reduce tobacco consumption. Curr. Environ. Health Rep. **1**: 208–216. doi:10.1007/s40572-014-0016-x.

Novotny, T.E., Lum, K., Smith, E., Wang, V., and Barnes, R. 2009. Cigarettes butts and the case for an environmental policy on hazardous cigarette waste. Int. J. Environ. Res. Public Health, **6**: 1691–1705. doi:10.3390/ijerph6051691.

Ocean Conservancy. 2022. International Coastal Cleanup, 2019 Report. Available from https://oceanconservancy.org/trash-free-seas/international-coastal-cleanup/annual-data-release/ [accessed 25 June 2022].

Oliva, M., Marchi, L., Cuccaro, A., and Pretti, C. 2021. Bioassay-based ecotoxicological investigation on marine and freshwater impact of cigarette butt littering. Environ. Pollut. **288**: 117787. doi:10.1016/j.envpol.2021.117787.

Osuala, F.I., Abiodun, O.A., Igwo-Ezikpe, M.N., Kemabonta, K.A., and Otitoloju, A.A. 2017. Relative toxicity of cigarette butts leachate and usefulness of antioxidant biomarker activity in Nile tilapia *Oreochromis niloticus* (Trewavas, 1983). Ethio. J. Environ. Stud. Manag. **10**: 75–88. doi:10.4314/ejesm.v10i1.8.

Parker, T.T., and Rayburn, J. 2017. A comparison of electronic and traditional cigarette butt leachate on the development of *Xenopus laevis* embryos. Toxicol. Rep. **4**: 77–82. doi:10.1016/j.toxrep.2017.01.003.

Poppendieck, D., Gong, M., and Pham, V. 2020. Influence of temperature, relative humidity, and water saturation on airborne emissions from cigarette butts. Sci. Total Environ. **712**: 136422. doi:10.1016/j.scitotenv.2019.136422.

Rahman, M.T., Mohajerani, A., and Giustozzi, F. 2020. Possible recycling of cigarette butts as fiber modifier in bitumen for asphalt concrete. Materials, **13**: 734. doi:10.3390/ma13030734.

Ramos, B., Alencar, M.V., Rodrigues, F.L., Lacerda, A.L.F., and Proietti, M.C. 2021. Spatio-temporal characterization of litter at a touristic sandy beach in South Brazil. Environ. Pollut. **280**: 116927. doi:10.1016/j.envpol.2021.116927.

Reinert, M. 1987. Classification Descendante Hiérarchique et Analyse Lexicale par Contexte-Application Au Corpus des Poésies D'A. Rimbaud. Bull. Sociol. Methodol. **13**: 53–90. doi:10.1177/075910638701300107.

Ribeiro, V.V., Lopes, T.C., dos Santos Pinto, M.A., Póvoa, A.A., Corrêa, V.R., De-la-Torre, G.E., et al. 2022. Cigarette butts in two urban areas from Brazil: links among environmental impacts, demography and market. Environ. Res. **213**: 113730. doi:10.1016/j.envres.2022.113730.

Ribeiro, V.V., Pinto, M.A., Mesquita, R.K., Moreira, L.B., Costa, M.F., and Castro, Í.B. 2021. Marine litter on a highly urbanized beach at Southeast Brazil: a contribution to the development of litter monitoring programs. Mar. Pollut. Bull. **163**: 111978. doi:10.1016/j.marpolbul.2021.111978.

Santos-Echeandía, J., Zéler, A., Gago, J., and Lacroix, C. 2021. The role of cigarette butts as vectors of metals in the marine environment: could it cause bioaccumulation in oysters? J. Hazard. Mater. **416**: 125816. doi:10.1016/j.jhazmat.2021.125816.

Selmar, D., Radwan, A., Abdalla, N., Taha, H., Wittke, C., El-Henawy, A., et al. 2018. Uptake of nicotine from discarded cigarette butts–a so far unconsidered path of contamination of plant-derived commodities. Environ. Pollut. **238**: 972–976. doi:10.1016/j.envpol.2018.01.113.

Shen, M., Li, Y., Song, B., Zhou, C., Gong, J., and Zeng, G. 2021. Smoked cigarette butts: Unignorable source for environmental microplastic fibers. Science of the Total Environment **791**: 148384. doi:10.1016/j.scitotenv.2021.148384.

Silva, J.S., Barbosa, S.C.T., and Costa, M. 2008. Flag items as a tool for monitoring solid wastes from users on urban beaches. J. Coast. Res. **24**: 890–898. doi:10.2112/06-0695.1

Slaughter, E., Gersberg, R.M., Watanabe, K., Rudolph, J., Stransky, C., and Novotny, T.E. 2011. Toxicity of cigarette butts, and their chemical components, to marine and freshwater fish. Tobacco Cont. **20**: 25–29. doi:10.1136/tc.2010.040170.

Smith, E.A., and Novotny, T.E. 2011. Whose butt is it? Tobacco industry research about smokers and cigarette butt waste. Tobacco Cont. **20**(Suppl 1): i2–i9. doi:10.1136/tc.2010.040105.

Soleimani, F., Dobaradaran, S., De-la-Torre, G.E., Schmidt, T.C., and Saeedi, R. 2021. Content of toxic components of cigarette, cigarette smoke vs cigarette butts: a comprehensive systematic review. Sci. Total Environ. **813**:152667. doi:10.1016/j.scitotenv.2021.152667.

Tehran Times. 2022b. Iran's 5-month tobacco exports rise 10 folds. Available from https://www.tehrantimes.com/news/441274/Iran-s-5-month-tobacco-exports-rise-10-folds [accessed 25 June 2022].

Thiel, M., Lorca, B.B., Bravo, L., Hinojosa, I.A., and Meneses, H.Z. 2021. Daily accumulation rates of marine litter on the shores of Rapa Nui (Easter Island) in the South Pacific Ocean. Mar. Pollut. Bull. **169**: 112535. doi:10.1016/j.marpolbul.2021.112535.

Torkashvand, J., Godini, K., Jafari, A.J., Esrafili, A., and Farzadkia, M. 2021. Assessment of littered cigarette butt in urban environment, using of new cigarette butt pollution index (CBPI). Sci. Total Environ. **769**: 144864. doi:10.1016/j.scitotenv.2020.144864.

Valiente, R., Escobar, F., Pearce, J., Bilal, U., Franco, M., and Sureda, X. 2020. Estimating and mapping cigarette butt littering in urban environments: a GIS approach. Environ. Res. **183**:109142. doi:10.1016/j.envres.2020.109142.

Venugopal, P.D., Hanna, S.K., Gagliano, G.G., and Chang, H.W. 2021. No butts on the beach: aquatic toxicity of cigarette butt leachate chemicals. Tob. Regul. Sci. **7**: 17–30. doi:10.18001/TRS.7.1.2.

Zoffoli, H.J.O., do Amaral-Sobrinho, N.M.B., Zonta, E., Luisi, M.V., Marcon, G., and Tolón-Becerra, A. 2013. Inputs of heavy metals due to agrochemical use in tobacco fields in Brazil's Southern Region. Environ. Monit. Assess. **185**: 2423–2437. doi:10.1007/s10661-012-2721-y.

Environ. Rev. Downloaded from cdnsciencepub.com by 179.240.18.129 on 02/10/23
For personal use only.